IS IN-CONTEXT LEARNING SUFFICIENT FOR INSTRUCTION FOLLOWING IN LLMS?

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ABSTRACT

In-context learning (ICL) allows LLMs to learn from examples without changing their weights: this is a particularly promising capability for *long-context* LLMs that can potentially learn from *many* examples. Recently, Lin et al. (2024) proposed URIAL, a method using only three in-context examples to align base LLMs, achieving non-trivial instruction following performance. In this work, we show that, while effective, ICL alignment with URIAL still underperforms compared to instruction fine-tuning on the established benchmark MT-Bench, especially with more capable base LLMs. We then uncover the most relevant elements for successful in-context alignment, finding the crucial role of the decoding parameters. Based on these insights, we show that the approach of URIAL can indeed be improved by adding *high-quality*, possibly carefully selected via greedy search, demonstrations in context, getting closer to the performance of instruct models. Finally, we provide the first, to our knowledge, systematic comparison of ICL and instruction fine-tuning (IFT) for instruction following in the low data regime, where ICL can be a viable alternative to IFT. Overall, our work advances the understanding of ICL as an alignment technique and its relationship to IFT. We provide our code at https://github.com/tml-epfl/icl-alignment.

1 INTRODUCTION

The large-scale pre-training phase allows Large Language Models (LLMs) to acquire extensive knowledge and capabilities (Bubeck et al., 2023). However, these base models struggle to follow instructions directly from prompts, necessitating further fine-tuning to be used as conversational models. Traditional alignment methods include Supervised Fine-Tuning (SFT) on instruction datasets (IFT, Wei et al., 2021; Chung et al., 2022) or preference data (DPO, KTO, IPO, and ORPO, Rafailov et al., 2023; Ethayarajh et al., 2024; Azar et al., 2024; Hong et al., 2024), and reinforcement learning (RLHF and RLAIF, Ouyang et al., 2022; Bai et al., 2022). These approaches typically involve multiple rounds of refinement of the instructed model (Touvron et al., 2023), i.e. high computational cost, and need a large amount of annotated data like human preferences, which might be difficult to collect. However, a line of work (Taori et al., 2023; Chen et al., 2023; Zhou et al., 2023; Lee et al., 2023; Zhao et al., 2024; Kaur et al., 2024) has suggested that IFT on a small amount of high quality instruction-following examples, even only 1000, can be sufficient to match the performance of more complex alignment mechanisms. In particular, Zhou et al. (2023) introduced the Superficial Alignment Hypothesis, stating that LLMs acquire all their capabilities during pre-training, and fine-tuning only allows the models to better access such knowledge when interacting with users (Gudibande et al., 2023; Duan et al., 2023). Using such small instruction datasets for IFT has the clear advantage of significantly reducing the cost of model alignment.

Inspired by Brown et al. (2020) who showed that LLMs can learn from demonstrations provided as part of the input—a concept known as in-context learning (ICL)—Lin et al. (2024) have recently studied the feasibility of *in-context alignment* (Han, 2023; Li et al., 2023b). They found that including merely three carefully selected question-answer pairs in the prompt is sufficient to make *base* models follow instructions and interact with users at a similar level to instruction-fine-tuned models on their own benchmark. This fact strongly validates the idea that alignment can be achieved at low computational cost and with few examples, as well the Superficial Alignment Hypothesis, as it does not even require to modify the model weights. Moreover, in-context alignment is especially appealing since it opens up the possibility of customizing base models via ICL, i.e., without any fine-tuning, which is potentially game-changing due to its simplicity and flexibility. While the results of Lin et al. (2024) are promising, they are restricted to a small number of base models and their own instruction following dataset.

Analysis of in-context alignment. In this work, we extend the evaluation of the URIAL (the abbreviation of Untuned LLMs with Restyled In-context ALignment) prompt strategy proposed by Lin et al. (2024) across several base models, including GPT-4-Base,[1] and on established instruction following benchmark MT-Bench (Zheng et al., 2023), see Table 1. First, we show that, although URIAL achieves reasonable performance, it still lags behind instruction-fine-tuned models. Then, to better understand the success but also weaknesses of in-context alignment, we analyze which are the most relevant ingredients in URIAL. We find that the decoding parameters in the LLMs generation (temperature, sampling scheme, etc.) may crucially influence the performance, and the optimal configuration allows even base models to achieve good scores on MT-Bench. Moreover, we show that randomly sampled triplets of examples from the highly curated instruction dataset SkillMix (Kaur et al., 2024) can achieve similar, or even better, results than URIAL when used for in-context alignment.

Table 1: **Systematic comparison of URIAL to aligned models on MT-Bench across different base LLMs.** For several recent model families, we compare the performance on instruction following tasks of the base LLMs plus URIAL (i.e. in-context alignment) to that of the instruct models, fine-tuned with sophisticated techniques like supervised instruction fine-tuning and RLHF. In most cases, the fine-tuned models outperform URIAL. * denotes the result taken from the URIAL GitHub repository.

Model	1st-turn	2nd-turn	Average
Llama-2-7B + URIAL *	5.75	3.91	4.83
Llama-2-7B-Instruct	**7.14**	**5.91**	**6.53**
Llama-2-70B + URIAL *	**7.61**	6.61	7.11
Llama-2-70B-Instruct	7.37	**7.03**	**7.20**
Llama-3-8B + URIAL *	6.84	4.65	5.75
Llama-3-8B-Instruct	**8.29**	**7.42**	**7.86**
Llama-3-70B + URIAL *	7.71	5.09	6.40
Llama-3-70B-Instruct	**8.96**	**8.51**	**8.74**
Llama-3.1-8B + URIAL *	6.95	5.31	6.13
Llama-3.1-8B-Instruct	**8.27**	**7.73**	**8.00**
Mistral-7B-v0.1 + URIAL *	**7.49**	5.86	6.67
Mistral-7B-Instruct-v0.1	7.31	**6.39**	**6.85**
Mistral-7B-v0.2 + URIAL *	6.99	5.55	6.27
Mistral-7B-Instruct-v0.2	**8.06**	**7.21**	**7.64**
Mixtral-8x22B-v0.1-4bit + URIAL	8.28	7.14	7.71
Mixtral-8x22B-Instruct-v0.1-4bit	**8.78**	**8.25**	**8.52**
GPT-4-Base + URIAL	7.96	5.04	6.50
GPT-4 (March 2023) *	**8.96**	**9.03**	**8.99**

Many-shot in-context alignment. Then, we test various strategies to improve in-context alignment, leveraging recent models with *extensive context windows* which allow for longer in-context prompts. In particular, we study the effect of *many-shot* in-context learning by adding *high-quality* demonstrations from existing instruction datasets. Unlike what suggested by Lin et al. (2024), this approach can improve upon URIAL, but only when using high-quality examples, as those from SkillMix. However, it is still not sufficient to fully close the gap with aligned LLMs, as the performance plateaus after 10-30 in-context examples. This behavior is in contrast to many-shot ICL for tasks like summarization (Narayan et al., 2018), translation (Costa-jussà et al., 2022), or classification (Li et al., 2024), where providing many examples is clearly beneficial (Agarwal et al., 2024; Bertsch et al., 2024). We further test a simple greedy algorithm to select the in-context examples which optimize the MT-Bench score. This selection scheme outperforms, with 1 to 3 additional demonstrations, the many-shot approach with random samples, and allows to further reduce the distance from in-context aligned models to fine-tuned models.

ICL vs IFT for alignment. While our experiments suggest that ICL cannot match the instruction-following performance of models aligned through costly fine-tuning, possibly using preference data, it remains an open question whether ICL can compete with or outperform IFT in the low-data regime. We provide an extensive evaluation on multiple LLMs and datasets of both approaches when varying the question-answer dataset size between 3 and 4000 examples. We observe that, with high-quality data, ICL and IFT achieve almost identical 1st-turn MT-Bench score. Surprisingly, in the 2nd-turn score, IFT clearly outperforms ICL, with ICL performing even worse than the base model. This suggests that ICL overfits to the style of the examples shown in context.

[1]We received access to the base GPT-4 model via the OpenAI Researcher Access Program.

Overall, these results, complemented by several ablation studies, provide a more nuanced picture of ICL as an alignment technique compared to previous works. Moreover, our comparison of ICL to IFT when using the same data bridges a gap in the literature about understanding these orthogonal approaches for adapting pre-trained LLMs into conversational models. In summary, our contributions are as follows,

- **Analysis of In-Context Alignment** (Sec. 2): We systematically evaluate URIAL on a broader set of base LLMs, including GPT-4-Base, with established instruction-following benchmark. Our results indicate that in-context alignment with URIAL still underperforms instruction fine-tuning and more sophisticated alignment methods, and a proper decoding scheme is the crucial ingredient behind the empirical success of URIAL.
- **Scaling In-Context Alignment** (Sec. 3): We find that many-shot ICL with high-quality examples and a simple greedy algorithm can reduce the gap between in-context aligned models to fine-tuned models.
- **First systematic comparison of ICL vs IFT for instruction following** (Sec. 4): We show that ICL and IFT with the same number of examples are roughly equivalent for single-turn conversations in the low-data regime. However, IFT generalizes substantially better than ICL when more examples are present, especially for multi-turn conversations.

2 UNCOVERING THE LIMITS OF ICL FOR INSTRUCTION FOLLOWING

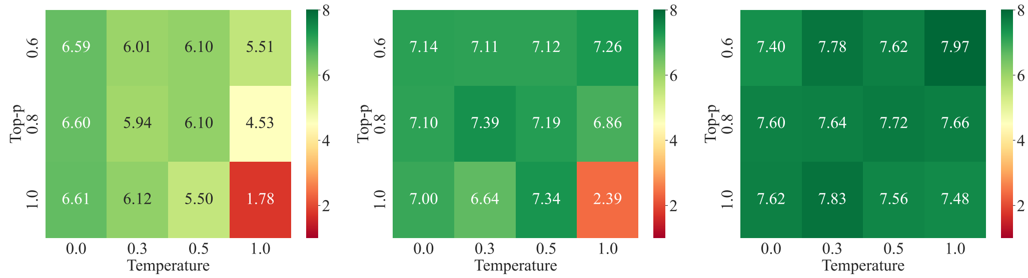
In the following, we provide an in-depth analysis which aims at (1) systematically comparing the performance of base models plus URIAL to that of aligned models (Sec. 2.1), (2) understanding which components of URIAL (and in-context alignment in general) are most important for its effectiveness (Sec. 2.2), (3) testing the influence of question-answers format in our task (Sec. 2.3).

2.1 SYSTEMATIC EVALUATION OF URIAL

Background on URIAL. Lin et al. (2024) propose to use a prompt with as few as three constant stylistic examples, which follow a curated answering structure that human users are familiar with, and a carefully designed set of rules, highlighting the format and principles AI assistant needs to follow strictly when generating answers, to align base LLMs with ICL. We present the complete URIAL prompt in Fig. 9. The highly curated examples always begin with affirming the user query by saying something like “Hi, I’m happy to help you.”, then proceed to answer the query by enumerating necessary items and steps, and conclude with an engaging summary. The set of rules first elucidate the scenario and format of the subsequent interaction between the human user and the AI assistant, then outline multiple qualitative aspects, including helpfulness, honesty, and harmlessness, which the base LLMs should adhere to when answering queries.

Experimental setup. We compare the performance of several base models with the URIAL in-context prompt to that of their instruction fine-tuned versions. Table 1 shows the results on MT-Bench (Zheng et al., 2023), one of the most popular benchmarks for instruction following ability of LLMs. We note that Lin et al. (2024) originally tested URIAL on their own dataset, in which only 8% of the examples are obtained from MT-Bench. We compare several LLMs of different sizes and capabilities, ranging from Llama-2-7B (Touvron et al., 2023) to the *base* GPT-4 model (OpenAI, 2023). For simplicity, we consider the *official* instruction-tuned LLMs, i.e., provided from the same source of the base models, even if there exist third-party fine-tuned versions which can achieve higher scores.

Results. In Table 1 we observe that base models with URIAL achieve competitive performance on the 1st-turn score, but cannot match that of their instruction fine-tuned counterparts in all cases except for Llama-2-70B and Mistral-7B-v0.1 (both originally included in Lin et al. (2024), unlike most others). Conversely, the 2nd-turn score with URIAL is *significantly worse* than for instruction-tuned LLMs: we hypothesize that this is because no multi-turn demonstrations are given in context. Indeed, Zhou et al. (2023) show that, even in IFT, having a few multi-turn training examples significantly improve the performance of the model on multi-turn conversations. Because of this fact, in the rest of the paper, we mainly focus on the 1st-turn score and single-round conversations, and track of 2nd-turn performance for completeness. To fill in the gap in the influence of multi-turn conversations, we study multi-turn alignment via ICL in Sec. 2.4. Finally, we detail the breakdown of the results over categories in Table 11 (see App. E.4 for details).



(a) Base model (b) Base model with URIAL (c) Instruct model

Figure 1: **Effect of decoding parameters on the 1st-turn MT-Bench scores across models.** We vary temperature and top-p, with a fixed repetition penalty of 1.15 as used in the URIAL codebase. The heatmaps show that the answering quality of the base model (Mistral-7B-v0.2) with and without URIAL is sensitive to the decoding schemes. Conversely, the performance of the instruct model is robust to varying the decoding parameters. Surprisingly, with proper decoding parameters, the base model alone is already capable of following instructions. See the complete results in App. E.3.

2.2 WHAT MATTERS FOR IN-CONTEXT ALIGNMENT VIA URIAL?

Decoding parameters. When computing the performance of URIAL, Lin et al. (2024) use decoding parameters (temperature $= 0$, top-$p = 1$, repetition penalty $= 1.15$) which differ from the default ones in MT-Bench (where temperature depends on the topic of the question, top-$p = 1$, repetition penalty $= 1$). To clarify the influence of these parameters that were not explicitly discussed in Lin et al. (2024), we compute the performance of base LLM, base LLM with URIAL, and fine-tuned LLM when varying decoding configurations. In Fig. 1 we fix the repetition penalty to 1.15, and vary temperature and top-p, for the Mistral-7B-v0.2 models. Surprisingly, we notice that with the decoding parameters of URIAL (temperature $= 0$, top-$p = 1$) even the base model without any in-context example achieves reasonable MT-Bench score (6.61 vs 7.00 of URIAL). Moreover, the temperature value appears to have the most influence of the performance of the base model. With URIAL, almost all configurations provide very similar results, with the exception of temperature $= 1$, top-$p = 1$ (possibly because it allows the sampling of generated tokens from the tail of the token distribution, thus producing low-quality text). Finally, the aligned model performs similarly across all decoding schemes, with slightly better results than URIAL. These results suggest that (1) the decoding parameters are an *overlooked factor* contributing to the success of in-context alignment, and (2) fine-tuning adjusts the sampling distribution of language models so that even with high-variance decoding configurations the generated text preserves high quality. We provide additional analysis for no repetition penalty (i.e. $= 1$) in Fig. 13 and Llama-3.1-8B in Fig. 14 in App. E.3: these extensive experiments further validate, with different base LLMs and settings, that the decoding scheme is a crucial factor for the instruction-following behavior of base models, and also affects, but to lower degree, in-context alignment. Instead, the fine-tuned models are robust to variations in the decoding parameters. Then, for the rest of the experiments we fix the decoding scheme of URIAL.

In-context prompt. Next, we want to disentangle the influence of the various elements of URIAL on instruction following performance. In Fig. 2 we track the MT-Bench scores when putting in-context all possible subsets of the three demonstrations of URIAL (i.e. from 0 to 3 examples are used). Moreover, for each case we test either adding or not the part of the prompt including the rules to follow. First, we find that neither 1st-turn nor 2nd-turn MT-Bench scores are significantly influenced by the addition of the set of rules, in turn highlighting the importance of the examples. Focusing on 1st-turn score (left plot in Fig. 2), we see that increasing the number of in-context demonstration progressively improves the performance of the base model (Mistral-7B-v0.2 in this case): this observation motivates us to add further high-quality in-context demonstrations, see Sec. 3. Finally, the 2nd-turn has the opposite trend, getting degraded by more (single-round) demonstrations.

2.3 IMPORTANCE OF QUESTION-ANSWER MATCHING

Surprisingly, Min et al. (2022) showed that using demonstrations with *random labels* does not significantly impair the results of ICL on classification and multiple choice tasks. We conduct a similar study for instruction following, see Table 2. Denoting $\{(X_i, Y_i)\}_i$ the set of in-context examples,

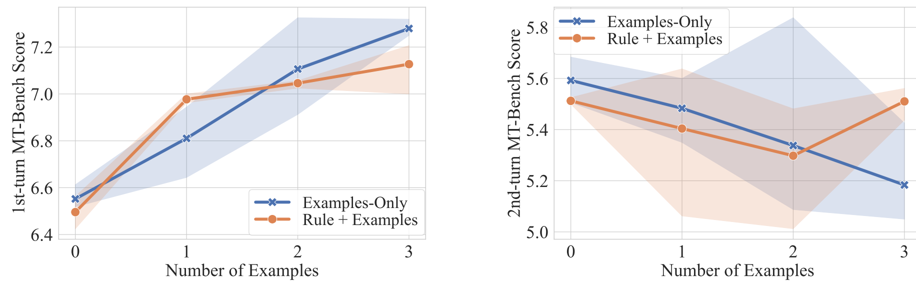


Figure 2: **Influence of the individual components of URIAL.** We report 1st and 2nd turn MT-Bench score for every subset of the three demonstrations of URIAL with Mistral-7B-v0.2. We test each configuration with ("Rule + Examples") and without ("Examples-Only") the URIAL set of rules in the in-context prompt. We observe a clear increasing trend in 1st-turn score with more examples, but a decrease in the 2nd-turn performance. The set of rules does not seem to influence the results. The randomness for 0 and 3 examples is caused by the small fluctuations in the score of the GPT-4 judge.

Table 2: **Importance of question-answer matching of demonstrations for in-context alignment.** We report the 1st-turn MT-Bench score of Mistral-7B-v0.2 and Llama-3.1-8B when varying the structure of the in-context examples $\{(X_i, Y_i)\}_i$, where X_i is the query and Y_i is the corresponding ground-truth answer.

In-context prompt	URIAL (3 examples)		URIAL + Greedy Search (6 examples)	
	Mistral-7B-v0.2	Llama-3.1-8B	Mistral-7B-v0.2	Llama-3.1-8B
no demonstrations	6.52	6.25	6.52	6.25
X only (no Y)	5.90	4.48	5.53	5.10
Y only (no X)	<u>6.94</u>	<u>6.83</u>	<u>7.02</u>	<u>6.98</u>
circular shift of Y	5.04	5.69	5.59	2.13
in-domain random Y	6.24	6.26	4.63	5.49
out-of-domain random Y	4.13	3.73	1.50	4.36
correct Y	**6.99**	**6.95**	**7.43**	**7.81**

with X_i the query and Y_i the corresponding ground-truth answer, we test several configurations on two sets of $\{(X_i, Y_i)\}_i$: the 3 URIAL examples and the 3 URIAL examples with the 3 examples found by our greedy search from Sec. 3.2 (to check the effect of increasing the number of demonstrations).

First, we do not use any demonstration, i.e., the original base models: with the decoding parameters found in the previous section, this already achieves reasonable results. Surprisingly, providing *the questions without the answers* (second row in Table 2) degrades the performance, while the opposite, *answers only*, is effective (0.4-0.7 higher scores than the base model). Next, we permute the answers Y_is with a circular shift of one position, so that all correct answers are still contained in the prompt but matched with the wrong question: this significantly degrades the performance, especially with more examples (URIAL + Greedy Search), e.g. Llama-3.1-8B attains a score of only 2.13 (note that the minimum score is 1). Also, for each question, we sample a new answer from those provided for other instructions in the same category (*in-domain*): e.g., a question about coding is paired with an answer from a different coding question, ensuring that, while the content may be incorrect, the style remains appropriate. Although worse than the original one, this configuration achieves decent scores. Finally, we sample answers from instructions belonging to different *out-of-domain* categories, so that even their style does not match what expected for each question: this leads to the worst performance.

These results show that not all the conclusions from Min et al. (2022) apply to ICL used for instruction following. Most importantly, using answers with correct content and, especially, correct style is crucial for the success of ICL. This property becomes even more evident when increasing the number of examples (to 6 instead of 3), and motivates us to use a highly-curated instruction dataset like SkillMix in experiments in the next sections. Finally, this result suggests that ICL for instruction following works differently compared to other less open-ended tasks.

2.4 MULTI-TURN ALIGNMENT VIA ICL

In IFT, Zhou et al. (2023) find that adding a few multi-turn examples to the training data significantly improves the model's performance on multi-turn conversations. URIAL (Lin et al., 2024) shows that